INCH POUND

MIL-B-44426B
22 February 1995
SUPERSEDING
MIL-B-44426A
26 June 1992

MILITARY SPECIFICATION

BOOTS, INTERMEDIATE COLD/WET

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers black, waterproof, mildew-resistant, insulated combat boots with a closed loop speed lace system and a welt sole for wear in intermediate cold/wet climates.

1.2 Classification. The boots shall be of one type in the following sizes and widths as specified (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FQSE, 2800 South 20th Street, Philadelphia, PA 19145-0599, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8430

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Schedule of sizes

Width 1/	3	3-1/2	4	4-1/2	5	5-1/2	6	6-1/2	7	7-1/2	8	8-1/2
N	X	X	X	X	X	X	X	X	X	X	X	X
R	X	X	X	X	X	X	X	X	X	X	X	X
W	X	X	X	X	X	X	X	X	X	X	X	X
XW	X	X	X	X	X	X	X	X	X	X	X	X

Width 1/	9	9-1/2	10	10-1/2	11	11-1/2	12	12-1/2	13	13-1/2	14
N	X	X	X	X	X	X	X	X	X	X	X
R	X	X	X	X	X	X	X	X	X	X	X
W	X	X	X	X	X	X	X	X	X	X	X
XW	X	X	X	X	X	X	X	X	X	X	X

1/ N = B, R= D, W =EE, and XW = EEE

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-203	- Paper, Kraft, Wrapping
A-A-1249	- Paper, Wrapping, Tissue
V-L-61	- Laces, Nylon
V-T-285	- Thread, Polyester
V-T-295	- Thread, Nylon
KK-W-231	- Welting, Leather, Shoe
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock

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MIL-A-8243	- Anti-icing and Deicing-Defrosting Fluids
MIL-L-3122	- Leather, Cattlehide, for Footwear Uppers, Chrome Tanned, Fatliquored
MIL-T-5624	- Turbine Fuel, Aviation Grades JP-4, JP-5, and JP-5/JP-8 ST
MIL-L-10867	- Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-41814	- Counter, Footwear
MIL-C- 43956	- Cloth, Twill, Cotton, 10.0 ounces

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-311	- Leather, Methods of Sampling and Testing
FED-STD-601	- Rubber; Sampling and Testing

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-731	- Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-1-1635	-	Speed Lace Assembly

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017).

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RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580-0001).

U.S. Postal Manual

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Chromatic Transference Scale

Gray Scale for Color Change

AATCC 8 - Colorfastness to Crocking
AATCC 15 - Colorfastness to Perspiration
AATCC 107 - Colorfastness to Water

(Applications for copies should be addressed to the AATCC National Headquarters, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 573 - Rubber Deterioration in an Air Oven
D 624 - Rubber Property - Resistance
D 1565 - Specification for Flexible Cellular Materials - Vinyl Chloride Polymers and Copolymers (Open Cell Foam)
D 1630 - Rubber Property - Abrasion Resistance (NBS Abraded)
D 2240 - Rubber Property - Durometer Hardness
D 3574 - Method of Testing Flexible Cellular Materials - Slab, Bonded and Molded Urethane Foams
E 96 - Water Vapor Transmission of Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. The use of recycled materials is encouraged when practicable as long as it meets the requirements of this specification.

3.2.1 Leather

3.2.1.1 Upper leather vamps, quarters, eyelet stays, and backstays. The upper leather shall conform to treatment A and treatment B of MIL-L-3122 with a mellow temper, except that the leather shall be full grain only. The eyelet stays shall be thinner pliable portions of leather split to a thickness of not less than 2-1/2 ounces nor more than 3-1/2 ounces. The leather shall be drum-dyed black with a grain surface finish that, when scuffed, may be restored to uniformity by simply brushing or applying commercial polish containing silicone. The leather shall be finished so that it has an adequate moisture vapor transmission rate for breathable footwear. The leather shall conform to style Thule J74000 (see 6.5.1.1), style Boulder HH 13339 (see 6.5.1.2) or equal (see 6.7).

3.2.1.2 Tongues and collars. The tongues and collars shall be gusset leather conforming to treatment A and B of MIL-L-10867. The color shall be black. At least 80 percent of the eyelet stay specimens tested shall have a slot tear strength of not less than 30 pounds when tested in accordance with MIL-L-3122.

3.2.2 Insoles. The insole shall be cut from Texorist material (see 6.5.2). The Texorist shall contain 0.25 to 1.00 percent copper-8-quinolinolate uniformly distributed throughout the insole. The Texorist insole shall be 5-1/2 irons or 0.105 $\pm$ 0.003 inch thick. The insole shall be cut with the heel-to-toe direction across the machine direction of the Texorist. The Texorist insole shall have a stuck-on rib applied to the printed side of the material.

3.2.3 Fabrics

3.2.3.1 Bootie lining. The bootie lining shall be constructed from no more than three pattern pieces cut from a waterproofed five-layer laminated quilted package material (see 6.5.3).

3.2.3.1.1 Five-layer laminate. The five-layer laminate lining consists of a three-layer quilted package, a plastic film, and a tricot knit cloth conforming to the following requirements.

3.2.3.1.1.1 Quilted package. The three-layer quilted package (see 6.5.4) shall consist of the following:

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a. Top layer. The top layer shall consist of a thermal bonded Nylon 6,6. The weight shall be 4.3 ± 0.7 ounces per square yard. The thickness shall be 0.028 ± 0.008 inch.

b. Middle layer. The middle layer shall be Thinsulate B-200 or equal. As an alternate, Thermolite 200 TFI may be used.

c. Bottom layer. The bottom layer shall consist of a chemical and thermal bonded Nylon 6,6. The weight shall be 0.4 ± 0.1 ounces per square yard.

3.2.3.1.1.2 Plastic film. The plastic film shall be polytetrafluoroethylene, microporous. The weight shall be 0.3 ± 0.01 ounces per square yard.

3.2.3.1.1.3 Tricot knit. The tricot knit cloth shall be nylon. The weight shall be 0.8 ± 0.10 ounces per square yard. The color shall be neutral.

3.2.3.1.2 Vamp lining (alternate box toe). The vamp lining shall be cut from cotton twill cloth conforming to class 2 of MIL-C-43956.

3.2.3.2 Adhesive. The adhesive for laminating the layers together shall be such that the laminated cloth will meet the requirements of this specification. The adhesive shall have no adverse health hazard when used as intended.

3.2.3.3 Laminated cloth. The quilted package specified above shall be laminated to one side of the plastic film and the tricot knit cloth shall be laminated to the other side of the plastic film. The face side of the laminated cloth shall be the side with the quilted package. The laminated cloth shall conform to the requirements in Table I when tested as specified in 4.4.1.

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TABLE I. Physical requirements

Characteristics	Minimum	Maximum
Weight, oz./sq. yd.	11.00	15.00
Thickness (inches)	0.070	0.170
Thermal shrinkage Machine (%):		
Sample mean		-2.40
Lot mean	1.16	-2.35
Thermal shrinkage Transverse (%)		
Sample mean	-	-2.40
Lot mean		
Moisture vapor transmission rate (g/m^2/24 hrs):		
Procedure B	640	-
Procedure BW	2770	-
Water permeability		
Initial	no leakage	

3.2.3.3.1 Color. The color of the face side of the laminated cloth shall be a solid Nicotine and the color of the back side, or tricot knit side, shall be natural. Both sides of the cloth shall match the representative sides of the standard sample.

3.2.3.3.2 Matching. The color shall match the standard sample (see 6.4) when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7500 ± 200K with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K.

3.2.3.3.3 Colorfastness. The face side of the laminated cloth shall meet the colorfastness requirements specified below when tested in accordance with 4.4.1. Test one specimen per sample. The light equipment shall be Macbeth Model #SPL-75, or equal (see 6.7).

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Characteristic	Colorfastness requirements
Fastness to water: 1/	
Color change 2/	4 minimum
Staining 3/	4 minimum
Fastness to perspiration: 1/	
Color change 2/	4 minimum
Staining 3/	3-4 minimum
Fastness to crocking:	
Dry 3/	3-4 minimum
Wet 3/	3-4 minimum

1/ Remove the top layer from the laminated cloth specimens before testing.

2/ Evaluate color change using the AATCC Gray scale for Color Change.

3/ Evaluate staining using AATCC Chromatic Transference Scale. Color-fastness requirements are based on the lowest rating given any fiber on multifiber test fabric number 10.

3.2.4 Tapes

3.2.4.1 Seam sealing tape for bootie. The tape for sealing all designated seams and stitching in the bootie shall be cut 7/8 + 1/16, - 0 inch wide strips from natural colored laminated cloth fabricated in layers as follows (see 6.5.5.1):

a. A nylon tricot knit cloth weighing 0.80 ± 0.10 ounces per square yard.

b. The middle layer shall conform to the plastic film specified in 3.2.3.1.1.2.

c. Adhesive, a minimum of 5.4 mils in thickness.

3.2.4.2 Seam sealing tape for heel and counter area of bootie. The tape used to apply to the heel and counter area shall be a flexible two-layer film tape (see 6.5.5.2). It shall be designed to eliminate the tape blocking during application, and to impart strength and abrasion resistance to the sealed seams. It shall combine a low melting polyester adhesive with a very flexible polyurethane carrier layer. The tape shall be fabricated in layers as follows and shall meet the following requirements when tested as specified in 4.4.1.

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a. Adhesive layer. The adhesive layer shall be polyester. The thickness shall be 0.004 inch. The color shall be cloudy, translucent with a slight amber cast.

b. Carrier layer. The carrier layer shall be an ester-based polyurethane elastomer. The thickness shall be 0.002 inch. The color shall be hazy with an amber cast.

3.2.5 Collar foam. The foam for the collar shall be 3/8 ± 1/32 inch polyurethane open-cell foam having a density of 2 pounds per cubic foot.

3.2.6 Thermal quarter insulation. The thermal insulation shall be Thinsulate B-400 or equal. As an alternate, Thermolite 400 TFI may be used.

3.2.7 Stuck on rib. The stuck on rib shall go completely around the insole, starting and stopping on the inside or outside shank area.

3.2.7.1 Rib construction. The ribbing fabric used in the construction of the rib shall be unbleached cotton or cotton-synthetic fabric conforming to the requirements of table II when tested as specified in 4.4.1. The fabric shall be coated on one side with the rubber-base adhesive and bonded to the printed side of the insole. The fabric or fiberboard shall cover 5/32 -2/32, +1/32 inch of the peripheral edge and extend over the top of the vertical rib, or the fabric may extend from the inside top of the vertical rib and cover the area between the ribs for a distance not less than 5/8 inch from the inside rib edge. The preformed rib shall be firmly and smoothly fitted and bonded to the flesh side of the insole as specified above to provide for the required edge extension on the finished boot.

TABLE II. Ribbing fabric requirements

Characteristic	Requirement, minimum
Weight	8.0 oz./sq. yd.
Yarns per unit length:	
Warp	52 yarns per inch
Filling	30 yarns per inch
Breaking strength:	
Warp	105 pounds
Filling	105 pounds

3.2.7.2 Rib strength. The physical requirements for the rib strength shall conform to Table III when tested as specified in 4.4.1.

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TABLE III. Rib strength requirements

Characteristic	Minimum 1/	Average
Shear strength	70 pounds	75 pounds
Stitch strength	20 pounds	30 pounds

1/ No single determination (sample unit) shall fall below the minimum value specified and the average of all specimens (lot average) shall be not less than the average specified.

3.2.8 Box toe. The box toe shall be of virgin polyethylene, natural and unfilled (see 6.5.6). The specific gravity shall be 0.942 and the melt flow shall be 11.5. The thickness shall be an average of 0.045 to 0.055 inch. As an alternate, the box toe material shall be a laminate of 0.020 ± 0.002 inch virgin ionomer resin (surlyn) sandwiched between two layers of nonwoven polyester saturated with polystyrene. The total thickness shall be 0.056 ± 0.005 inch. The box toes shall be skived 3/8 ± 1/8 inch wide at the breast.

3.2.8.1 Box toe sizes. Box toes shall be made in seven sizes to cover the range of boot sizes and widths specified in Table IV. The box toes shall be fitted in accordance with Table IV. When the alternate box toe is used, the box toes shall be made in a sufficient number of sizes to cover boot sizes 3 through 14 in the necessary widths.

TABLE IV. Box toe fitting schedule

Boot toe size	Boot width and sizes			
	N	R	W	XW
2	3 - 3-1/2	3	----------	----------
4	4 - 5-1/2	3 - 5-1/2	3 - 4-1/2	3-4
6	6 - 7-1/2	5-1/2 - 7	5 - 6-1/2	4-1/2 - 6
8	8 - 9-1/2	7-1/2 - 9	7 - 8-1/2	6-1/2 - 8
10	10 - 11	9-1/2 - 10	9 - 10-1/2	8-1/2 - 10
12	12 - 13-1/2	11-1/2 - 13	11 - 12-1/2	10-1/2 - 12
14	14	13-1/2 - 14	13 - 14	12-1/2 - 14

3.2.9 Counters. The counters shall conform to the requirements of MIL-C-41814. The counters shall be made in a sufficient number of sizes to cover boot sizes 3 through 14 and in the necessary widths. Counters shall conform to the specific last used.

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TABLE V. Counter fitting schedule

Counter size	Boot width and sizes			
	N	R	W	XW
000	3 - 4-1/2	----------	----------	---------
00	5 - 6-1/2	3 - 4-1/2	----------	---------
0	7 - 8-1/2	5 - 6-1/2	3 - 4-1/2	----------
1	9 - 10-1/2	7 - 8-1/2	5 - 6-1/2	3 - 4-1/2
2	11 - 12-1/2	9 - 10-1/2	7 - 8-1/2	5 - 6-1/2
3	13 - 14	11 - 12-1/2	9 - 10-1/2	7 - 8-1/2
4	----------	13 - 14	11 - 12-1/2	9 - 10-1/2
5	----------	----------	13 - 14	11 - 12
6	----------	----------	----------	13 - 14

3.2.10 Welting. The welting shall conform to the requirements of KK-W-231. As an alternate, the vinyl welting shall be of flat top design, bright black in color, and may be formed with or without storm welt of medium size bead. The welting shall be of sufficient width to ensure that the required extensions on the finished boots are met.

3.2.11 Fiberglass shank. The fiberglass shank shall consist of glass fibers impregnated with vinyl ester resins encased in a plastic sleeve or tube (see 6.5.7). The width of the uncured, encased fiberglass resins shall be 5/8 ± 1/32 inch. The shank shall be cured, bonded to the insole, and shall conform to the shape of the last. When fully cured, the shank (glass and resin area) shall be 5/8 ± 1/16 inch in width. The forward end of the shank shall be tapered for a minimum of 3/8 inch. Testing shall be as specified in 4.4.1.

3.2.11.1 Shank fitting sizes. The shank fitting shall conform to the schedule as specified in Table VI.

TABLE VI. Shank lengths (+ 1/8 inch)

Shank size	5	5-1/4	5-1/2	5-3/4	6	6-1/4
Boot sizes (all widths)	3-4	4-1/2-6	6-1/2-8	8-1/2-10	10-1/2-12	12-1/2-14

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3.2.11.2 Shankboard. The shankboard shall be cut from Bonfoam 300MD using the Government furnished patterns (see Table VII).

3.2.12 Metal

3.2.12.1 Staples. A staple lasting shoe wire described as follows shall be used:

Size	022A
Finish	Brassy
Package	70 pound fiber spools
Tensile	130,000 to 160,000 psi

Chemical analysis:

Carbon	0.15 to 0.20
Manganese	0.60 to 0.90
Phosphorus	0.04 maximum
Sulphur	0.05 maximum
AISI or SAE	C-1018

3.2.12.2 Eyelets. The eyelets shall be gloss finish, nylon-coated brass, X-long, conforming to size AA and shall have a dark oxide finish.

3.2.12.3 Speed lace assembly. The speed lace assembly shall be a two-piece construction consisting of gloss finish, nylon coated brass speed lacer and nylon coated alternate full tubular rivet in accordance with Drawing 2-1-1635. The color shall be black.

3.2.13 Bottom filler

3.2.13.1 Cold process bottom filler. The cold process bottom filler shall be spreadable without the use of heat. It shall consist of a mixture of ground cork and a suitable binder in the proportion of two parts cork to one part binder by volume. The cork granules shall be free from bark. When the spread filler is dry and set, it shall consist of four parts cork to one part binder. It shall not soften at less than 150°F and shall be water resistant and flexible. As an alternate, when applied by semi-automatic equipment or method, the cold process bottom filler shall consist of a mixture of ground cork and a suitable binder in the proportion by weight of one part cork to five parts binder. Upon loss of solvent, it shall consist by weight of one part cork to 3-3/4 parts binder. The binder shall be water insoluble and have a softening point of no less than 125°F.

3.2.14 Rubber compounds

3.2.14.1 Midsole. The midsole shall be a 4 iron (1/12 inch) black neoprene.

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3.2.14.1.1 Cushion midsole. The cushion midsole shall be 10-1/2 iron ± 1/2 iron black Nuron crepe (see 6.10) or equal. (Ethylene Vinyl Acetate shall not be used).

3.2.14.1.2 Cushion heel. The cushion heel shall be 30 iron (5/8 inch) in thickness, black Nuron crepe or equal. (EVA shall not be used). The heel will have a bevel breast of 45° and the length of the bevel shall be 1 inch.

3.2.14.1.3 Rubber compound requirements

Nuron crepe:

Hardness	55 ± 10
Specific gravity	.50 ± .10
Stitch tear	12 ± 3 @ 12 iron
Shrinkage	3 percent max.

3.2.14.2 Outersole and heel. The outersole and heel shall be black rubber conforming to Vibram 148 Kletterlift Sole (see 6.5.9) or equal (see 6.7).

3.2.14.3 Rubber compound requirements. The requirements for the rubber are based on the testing done on test slabs (according to ASTM and FIA procedures) and are not actual soles. There may be some deviation on actual outersoles or midsoles based on style thickness and curing parameters. The requirements shall be as specified below.

Style	148 Kletterlift (outsole)	6 iron midsole
Compound	3389	330
NBS abrasion	150 min.	N/A
Hardness (unaged)	70-80A	N/A
Split tear	65 lb. min.	N/A
Stitch tear (Dry)	320 lb. min.	180 lb. min.
Stitch tear (Fuel B)	200 lb. min.	50 lb. min.
C tear	308 lb. min.	N/A
Ross flex, Oil #3	500% max., 100,000 flexes	N/A

3.2.14.4 Bond strength. The outersole to midsole bond strength shall be not less than 100 pounds when tested as specified in 4.4.4.

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3.2.15 Thread

3.2.15.1 Thread, upper fitting. The thread for all upper fitting operations shall conform to types I and II, class B (water repellent) of V-T-295. The color shall be black and the size shall be not less than E.

3.2.15.2 Thread, inseaming. The thread for all inseaming operations shall be polyester conforming to type I, class I, subclass B of V-T-285. The thread shall be size 10, 3 or 6 ply and natural in color.

3.2.15.3 Thread, sole stitching. The thread for sole stitching shall be polyester. The color of the running thread shall be black, and the color of the bobbin thread shall be natural. The bobbin thread shall be coated with wax.

3.2.16 Wax, inseam sewing and sole stitching. The wax shall be white or golden and shall be permanently plasticized resin. The wax, in the temperature range of normal machine use, shall thoroughly wax the threads described in 3.2.15.2 and 3.2.15.3.

3.2.17 Adhesives. The adhesives used for bonding the various leather and fabric components of the boots shall be of the following types:

a. Natural rubber latex
b. Synthetic rubber latex (including chloroprene)
c. Natural rubber solvent cement
d. Synthetic rubber solvent cement
e. Synthetic resin cement
f. Polyurethane rubber solvent cement
g. Thermoplastic hot melt

3.2.17.1 Outsole adhesive. The adhesive used for bonding the outsole, heel, cushion midsole and midsole shall produce a jet fuel oil resistant bond.

3.2.18 Finishing materials for leather uppers. Finishing materials used shall be compatible with the leather's silicone waterproofing agents and shall cause no adverse effect on water resistance of the leather of the boots.

3.2.18.1 Repairers. Repairers shall be liquid spray, crayon or paste type applied by hand. The color shall match the upper leather and have sufficient coverage to correct minor surface imperfections of the leather.

3.2.18.2 Fillers. Fillers, where used, shall be capable of providing a suitable foundation for the application of the top finish and shall be applied by sponge or spray method.

3.2.18.3 Top finish. The top finish shall be applied by either the sponge or spray method and shall be of a synthetic resin and wax base to even out the tannery finish but not to cover or change its natural appearance. It shall finish dry to a high gloss.

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3.2.19 Instruction tag. The instruction tag, to be tied to the finished boots, shall be printed on tag stock that is 5-1/4 by 6-1/2 inches and folded in the middle to form 5-1/4 by 3-1/4 inch printed pages in book form. A punch hole shall be made in the top near the folded edge for insertion of the lace. The instruction tag shall be printed using the data specified on Figure 1.

3.2.20 Molded cushion insole. The molded cushion insole shall be a medium brown microcellular urethane foam, with an integral skin laminated to a thermo-formed cross-link polyethylene heel cup/arch support. Before thermo-forming, the polyethylene shall have a density of 4 pounds per cubic foot ±10% and a minimum thickness of 0.200 inch. The polyurethane cover shall have a thickness of 0.110 ± 10 percent, and shall include an active organic anti-microbial agent (see 6.5.11). The total thickness of the insole (insert) shall be 0.155 to 0.195 inch forward of the instep arch and 0.190 to 0.215 inch elsewhere. The cushion insole shall have Cambrelle in nicotine color, flame laminated over the polyurethane cover. The polyurethane foam shall conform to the following properties:

Properties	Requirements
Density	17 + 2, -1 pound per cubic foot
Tensile strength	115 pounds per square inch (min.)
Compression set	10 percent (max.)
Compression set after hydrolysis	10 percent (max.)

1/ Less than 10 percent after 5 hours exposure to saturated steam at 15 psi.

The size of the cushion insole shall be that needed to cover the last bottom of the whole boot and shall be used in such size boot and the next larger half size (size 9R insole to be used in 9R and 9-1/2R). The insole shall be cut in such a manner that the full contoured portion of the heel area and arch support will be included.

3.2.20.1 Molded cushion insole (Alternate). As an alternate, the low density molded polyurethane heel cup arch support will be a polyurethane polyether system medium black in color molded directly to a Cambrelle top cloth, nicotine in color. The total thickness of the insole (insert) shall be 0.155 to 0.195 inch forward of the instep arch and 0.190 to 0.215 inch elsewhere. Each pair shall be moled on to the top cloth as blockers marked 0,1,2,3,4. Net cut sizes and widths will be cut from said blockers for placement into the product (see 6.5.11.1). The polyether polyurethane foam shall conform to the following properties:

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Properties	Requirements
Density	Shore "00" Hardness 60 = /_ 0.010, 17 + 2, - 1 lb/cubic foot
Tensile strength	82 (PSI)
Compression set (%)	(77°F) 25% @ 10.6 PSI 50% @ 21.6 PSI
Elongation (%)	132

3.2.21 Laces. The laces shall conform to type III, class 1 of V-L-61. The color of the laces shall be black. The length of each lace shall be a minimum of 72 inches for sizes 3 through 6-1/2, 87 inches for sizes 7 through 11, and 97 inches for sizes 11-1/2 through 14.

3.3 Design. The design shall be for a full lace closure with eyelet row edge of the quarters, grain-out leather upper, plain toe vamp, backstay, gusset tongue box toe, and three pattern piece bootie. The boot shall have a neoprene midsole and a full length rubber base outsole.

3.4 Pattern and dies, number 9218. The height of the finished boot, measured upward on the outside from the bottom at the breast of the heel to the top of the boot, shall be 11 $\pm$ 1/4 inch on size 9 and shall graduate up or down between whole sizes approximately 1/8 inch as indicated by patterns. A standard set of paper patterns will be furnished by the Government as a basis from which the contractor's dies or patterns shall be made. The patterns and contractor's cutting dies shall consist of the component parts as specified in Table VII. (The Government patterns shall not be altered in any way and shall be followed, except for lasting allowance, which shall be determined by the contractor). The contractor shall furnish insole paper patterns. When approved by the procuring activity, the bootie patterns may be altered to properly fit the last (see 3.5). The stitched uppers may be sized up or down 1/2 size and /or width, over/under the last size to accommodate variations in leather temper and to maintain proper lasting margins. The boots shall have their size markings adjusted accordingly.

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TABLE VII. Contractor's dies and patterns, number 9218

Component parts	Widths	Sizes
Vamps	4 widths	Whole and half sizes
Quarter	4 widths	Whole and half sizes
Tongue	4 widths	3 - 6-1/2, 7 - 8-1/2, 9 - 14, EEE 3 - 6-1/2, 7 - 9, 9-1/2 - 14, EE 3 - 6-1/2, 7 - 9-1/2, 10 - 14 D 3 - 6-1/2, 7 - 9-1/2, 10 - 14, B
Eyelet stay	4 widths	3 - 6-1/2, 7 -8-1/2, 9 - 14, EEE 3 - 6-1/2, 7 -9, 9-1/2 - 14, EE 3 - 6-1/2, 7 - 9-1/2, 10 - 14, D 3 - 6-1/2, 7 - 9-1/2, 10 - 14, B
Collar	1 width	3 - 6-1/2, 7 - 9-1/2, 10 - 11-1/2, 12 - 13-1/2, 14, B-D-EE-EEE
Backstay	4 widths	3 - 6-1/2, 7 - 8-1/2, 9 - 14, EEE 3 - 6-1/2, 7 - 9, 9-1/2 - 14, EE 3 - 6-1/2, 7 - 9-1/2, 10 - 14, D 3 - 6-1/2, 7 - 9-1/2, 10 - 14, B
Quarter filler	4 widths	Whole sizes
Collar foam	1 width	3 - 6-1/2, 7 - 9-1/2, 10 - 11-1/2, 12 - 13-1/2, 14, B-D-EE-EEE
Inside bootie	4 widths	Whole and half sizes
Outside bootie	4 widths	Whole and half sizes
Bootie bottom	4 widths	Whole and half sizes
Insoles	4 widths	Whole and half sizes
Shankboard	N/A	6 sizes

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TABLE VII. Contractor's dies and patterns, number 9218 (contd.)

Component parts	Widths	Sizes	
9218 Vamp marker line for tongue location and quarter front location:			
	3 - 6-1/2, D-EE-EEE	7 - 8-1/2, EEE	9 - 14, EEE
	3 - 6-1/2, B	7 - 9, EE	9-1/2 - 14, EE
		7 - 9-1/2, D	10 - 14, D
		7 - 9-1/2, B	10 - 14, B
9218 Quarter marker line for top of vamp location:			
	3 - 6-1/2, D-EE-EEE	7 - 8-1/2, EEE	9 - 14, EEE
	3 - 6-1/2, B	7 - 9, EE	9-1/2 - 14, EE
		7 - 9-1/2, D	10 - 14, D
		7 - 9-1/2, B	10 - 14, B
9218 Back tongue marker line for centering top of bootie lining and 6 stitches through front of tongue to hold bootie on center:			
	3 - 6-1/2, D-EE-EEE	7 - 8-1/2, EEE	9 - 14, EEE
	3 - 6-1/2, B	7 - 9, EE	9-1/2 - 14, EE
		7 - 9-1/2, D	10 - 14, D
		7 - 9-1/2, B	10 - 14, B
9218 Front tongue marker line for stitching single needle across front of tongue through the bootie lining:			
	3 - 6-1/2, D-EE-EEE	7 - 8-1/2, EEE	9 - 14, EEE
	3 - 6-1/2, B	7 - 9, EE	9-1/2 - 14, EE
		7 - 9-1/2, D	10 - 14, D
		7 - 9-1/2, B	10 - 14, B
9218 Eyelet stay marker line for boxstitching bottom of eyelet stay and tunnel hook location stab in cutting dies:			
	3 - 6-1/2, D-EE-EEE	7 - 8-1/2, EEE	9 - 14, EEE
	3 - 6-1/2, B	7 - 9, EE	9-1/2 - 14, EE
		7 - 9-1/2, D	10 - 14, D
		7 - 9-1/2, B	10 - 14, B

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TABLE VII. Contractor's dies and patterns, number 9218 (contd.)

Cutting instructions	Lasting instructions
Cut N up 1/2 on B	Last up 1/2 on M2673B-1 B
Cut R up 1/2 on D	Last up 1/2 on M2673B-1 D
Cut W up 1/2 on EE	Last up 1/2 on M2673B-1 EE
Cut XW up 1/2 on EEE	Last up 1/2 on M2673B-1 EEE

3.5 Lasts. The boots shall be made on model number M2673B-1 last (see 6.5.12.1), 701 last (see 6.5.12.2) or equal (see 6.7). A full set of patterns consisting of vamp, quarter, tongue, eyelet stay, backstay, bootie inside and outside, also bootie bottom and insoles will be loaned to the contractor by the Government.

3.6 Construction

3.6.1 Cutting uppers. The uppers shall be cut from grain out leather. The leather shall conform to the thickness specified in Table VIII; except for the eyelet stay, none of the upper parts shall be split or shaved. No parts cut off stretch shall be accepted. Vamps shall be cut from the bend area of the side leather.

TABLE VIII. Thickness of leather uppers

Part	Thickness (ounces)	
	Minimum	Maximum
Vamp	5	6
Quarter	4-1/2	6
Eyelet stay	2-1/2	3-1/2
Gusset	2-1/2	4

3.6.2 Skiving. The skiving of upper leather parts shall conform to the requirements specified in Table IX.

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TABLE IX. Skiving requirements

Part	Side	Location	Scarf width (inches)	Scarf edge thickness in millimeters (+ .3 mm)
Quarter	Flesh	Vamp location	3/4 ± 1/8	1.5
Quarter	Flesh	Eyestay location	3/4 ± 1/8	1.2
Vamp	Flesh	Lasting margin	3/4 ± 1/8	1.5
Vamp	Flesh	Rear top line	1/2 ± 1/8	1.3
Backstay	Flesh	Lasting margin	3/4 ± 1/8	1.5
Eyelet stay	Flesh	Top line	3/4 ± 1/8	1.1
Collar	Flesh	Inside Edge	3/4 ± 1/8	1.3

3.6.3 Marking, permanent identification. The inside quarter of each boot shall be marked on the grain side with the correct size and width, the contractor's identification symbol, and the month and year (expressed numerically) of the date of contract. This marking shall be impressed into the grain side of the leather in such a manner as to be permanently visible without cutting through at any point, and shall be placed approximately in the center of the inside quarter and between collar stitching. Figures shall be Arabic and letters shall be Gothic. The figures and letters shall be a minimum of 9/32 inch and a maximum of 3/8 inch in height. The contractor's symbol shall be in a block as shown by the following example: 10W/AB/ 6-92.

3.6.4 Tongue construction. The seam in the lower front of the tongue is lap seamed with the upper part lapped over to cover the stab in tongue. This is stitched on a single needle machine and bartacked on both ends. As an alternate (when approved by the procuring activity), a football shape cut out may be used in lieu of the overlap seam. This seam shall be closed inward using stitch types 301 or 401.

3.6.4.1 Vamp lining construction. The cloth vamp lining shall be designed to hold the box toe during the lasting process. The fabric shall cover the vamp from the throat edge forward to the vamp front edge and shall be cemented into place with an adhesive as specified in 3.2.17.

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3.6.5 Bootie lining construction. The lining material shall be cut with the diamond quilting stitching running long from the heel to toe. The lining consists of three cut pieces: outside quarter, inside quarter, and bottom. The thread for the construction of this bootie shall be white with a size 16 needle on the one needle operation and a size 20 needle on the union special closer. The bootie shall be closed using one or more of these closing methods: Singer lock stitch type 301, union special stitch type and zig zag stitch type. The bootie shall then be seamed sealed with the tape in 3.2.4.1. The second tape, as in 3.2.4.2, shall be applied to the heel area and extended up the back seam above the counter area. The bootie shall not show any leakage when tested for waterproofness as described in 4.4.1.

3.6.6 Upper leather fitting. Line marking patterns shall be used for all upper fitting. Inserts in cutting dies for stitch line impressions or die marking will not be allowed, except for the tongue and eyelet stay which will have stabs. Needles used for all fitting shall be no longer than size 18 except for Puritan vamping and automatic bartacking. All upper stitching shall be 8 to 10 stitches per inch, except Puritan vamping.

3.6.6.1 Vamping quarters to vamp. Vamping shall be done on a 3-N Puritan machine with three rows of stitching, 3/32 inch from the vamp top-line edge. Spacing between the three rows shall be 3/32. This upper stitching shall have 6 stitches per inch.

3.6.6.2 Eyelet stay to quarter. The eyelet stay shall be stitched to the upper with a single row stitch 3/32 from the back edge of the eyelet stay.

3.6.6.3 Closing back of upper. Quarters and vamp shall be closed at the back using overcasting stitch. The back seam shall be closed with the stitching closely positioned to the edge of the quarters and vamp on the grain side. The quarters and vamp shall then be rubbed open so that the quarters and vamp butt rather than lap. As an alternate, the quarters and vamp may be closed at the back by butting the edges, and using stitch type 502, 304, 404 or 503 with 10 to 12 stitches per inch. The back seam area shall be flat and smooth, and shall not contain any wrinkles or bunching. The back seam may be rubbed flat, if necessary.

3.6.6.4 Backstay fitting. The backstay fitting shall be stitched to the upper with two rows stitching 3/32 inch from outside edge of the backstay. There shall be a 3/32 inch space between the two rows of stitching. Excess backstay shall be trimmed off at the top edge of the backstay even with quarter.

3.6.6.5 Thermal insulation to quarter. An adhesive, specified in 3.2.17, shall be applied to the quarter insulation and then the quarter insulation shall be placed on the knit side of the bootie. As an alternate, the quarter insulation can be placed on the knit side of the bootie.

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3.6.6.6 Gusset tongue fitting. The gusset tongue shall be stitched to the vamp grain side out, with two rows of stitching, one row not more than 3/32 inch from the edge of the tongue, and a second row placed not more than 1/4 inch from the edge of the tongue. The tongue shall be stitched to the quarter eyelet stay at the top edge aligning nick-ins. The tongue shall then be stitched to the quarter eyerow with a stitch 3/32 inch from the outside edge and a row 3/32 inch from the existing eyerow stitch and trimmed flush with 1/8 inch trim.

3.6.6.7 Boxstitch eyelet stay. The eyelet stay shall be stitched to the vamp at the bottom of the eyelet stay. The stitch shall start in the middle at the bottom of the eyelet stay, 3/32 from edge, up the sides of the eyelet stay to marker line and across, down eyelet stay and across the bottom of the eyelet stay interlocking stitches at the bottom. This area shall then be pounded.

3.6.6.8 Bootie lining to upper fitting. The lining shall be placed inside the upper and stitched flush to the topline of the upper. The ends shall be placed under the gusset tongue and a bartack shall be added here for reinforcement. The lining shall then be tacked to front center of tongue, then stitched across front of tongue along marker line. Lining shall be trimmed at the top across tongue.

3.6.6.9 Foam padding and collar fitting. The collar shall be stitched flesh side out even with the top edge of the quarter with one row of stitching 3/32 inch from edge. The collar foam shall be cemented with an adhesive specified in 3.2.17, flush to the top edge of the lining, placed so that it extends to the inside eyerow stitching on each side of the gusset. The collar shall be turned over the padding and stitched with one row not more than 3/32 from the edge of the eyerow and shall end with a three-stitch back tack at the eyerow edge of the collar. Excess collar leather shall be cut off flush to the eyerow edges.

3.6.6.10 Eyeletting. The eyelets shall be equally spaced on each quarter. All sizes shall have two eyelets on each quarter.

3.6.6.11 Attaching speedlacer assembly. The speed lace loops shall be equally spaced on each quarter. All sizes shall have six speed lace loops on each quarter. The speed lace loops shall be perpendicular to the edge of the quarters and parallel to each other, and shall be securely riveted.

3.6.6.12 Lacing for lasting. The machine lacing or hand inserted clip(s) for lasting shall provide for an opening, of 7/8 inch.

3.6.6.13 Automatic bartacking. The bartack shall be done by the automatic method with tying of locking ends. When an automatic stitching machine is used, a size 20 needle may be used. There shall be two bartacks on each eyelet stay. The first bartack shall be below and parallel with the bottom topstitch line and the second shall be perpendicular with the outside box stitch line.

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3.6.7 Insole assembly. The stuck on rib is applied to the insole on the printed side, then the plain side is cemented and allowed to dry. The counter, which has been coated on the inside with a cement in 3.2.17 and fitted in accordance with Table V, shall then be stapled to the rib edge of the insole. There shall be five to seven staples to each counter. A cloth binding shall be stitched to the top edge of the counters before assembly.

3.6.8 Lasting. The correct size and widths of uppers, lasts, insoles and box toes shall be assembled. The upper/bootie shall be slipped on the last with the upper pulled up and away from the heel area. Nylon tape shall then be placed on the toe area for abrasion protection. Cement shall then be applied to the toe and heel area and the assembled insole then placed to the bootie and pressed to bond insole to bootie. Cement shall be put on the heel area and the upper pulled into place at the heel area. The preformed plastic toe, which has been coated on the inside with a cement, shall then be put in place. A last slip material shall be applied to the outside of the box toe for ease in pulling the leather during toe lasting. Uppers may be conditioned by any suitable means at toe lasting. Uppers shall then be drawn over the last with proper tension on the pulling-over machine. Cementing shall be done in conjunction with toe lasting, so that the toe is smoothly wiped-in and the toe staples attached securely around the toe at the base of the insole rib and the boxtoe. The sides shall then be pulled into place with hand pliers and the staples put around the shank and heel area through the base of the insole rib, counter, and upper. Excess upper shall then be trimmed at bottom to prepare for inseaming stitch.

3.6.8.1 Lasting (alternate box toe). The correct size and width of uppers, lasts, insoles, counters, and box toes shall be assembled. The upper bootie shall be slipped on the last with the upper pulled up and away from the heel area. Cement shall be applied to the heel area and the assembled insole then placed to the bootie and pressed to bond insole to bootie.

3.6.9 Inseaming. Inseaming shall be done by machine with the thread waxed and with proper tension on the thread, using a needle not larger than 41, and with not less than 3-1/4 stitches to the inch. The welting shall be inseam stitched to the bottom of the insole, rib end to rib end, and shall catch counter, box toe, and upper.

3.6.9.1 Inseam trimming. Excess upper, counter, and box toe shall be trimmed to the top of insole rib.

3.6.10 Shank fittings

3.6.10.1 Shank fitting-bottom filling. The shanks shall be selected for applicable size in accordance with the shank fitting schedule specified in 3.2.11.1. The shanks shall be inserted in position with the shank board cemented to the shank, filling the cavity between the edges of the vamp from the ball line rearward to the back of the heel seat. The forward end of the shank shall be flush with the insole and

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shall fit the contour of the shoe bottom back of the ball line. The bottom shall present a flat smooth surface for sole laying.

3.6.10.2 Shank fitting - bottom filling (alternate). An alternate method of bottom filling shall be to insert the shank into position and then the entire insole channel from toe to heel with bottom filler, thus eliminating the need for a shankboard. The bottom filler shall be firmly pressed into the insole channel to form a uniform even surface. The bottom shall present a flat smooth surface for sole laying.

3.6.11 Midsole assembly. A coating of adhesive shall be applied to both the neoprene and cushion midsoles and allowed ample time to dry. The midsoles shall be activated and spotted to each other and then pressed with sufficient pressure to insure a good bond. The combined midsole unit shall then be coated on the cushion midsole side in preparation for bonding to the welting.

3.6.12 Welting to midsole stitch. The welting and midsoles shall be stitched together on a lockstitch machine using waxed polyester thread using a number 47 needle and awl.

3.6.13 Sole laying. The outsole shall be roughed to remove any mold release and dirt from the surface. The outsole shall then be primed with an adhesive specified in 3.2.17 then cemented over the entire surface area. The neoprene side of the combined midsole unit shall then be wire brushed, then primed and cemented over the entire area and allowed ample time to dry. Both the outsole/heel combination and the midsole shall be placed in a midsole and be put on a sole laying machine with sufficient pressure to ensure a good bond. The soles, heels and combined midsole units shall be of adequate sole extension. The outside edges shall be pounded to secure the midsole to the midsole.

3.6.14 Rough rounding. Soles shall be uniformly rounded by any suitable means for the specified edge extension of sole and welt as specified in 3.6.17.

3.6.15 Sole toe stitching. Soles and welts shall be stitched together on a lockstitch machine using waxed polyester thread using a number 47 needle and awl. The stitch shall extend only around the toe area.

3.6.16 Pulling lasts. The lasting laces shall be cut from the uppers and lasts removed.

3.6.17 Edge trimming. Edges shall be trimmed using cutter #3x12-538KI-260 (12 tooth) or equal (see 6.7) to produce a rounded edge allowing for the required finished extension to be 3/16 $\pm$ 1/16 inch.

3.6.18 Sole finishing. Sole edge shall be trimmed square and smoothly scoured then stained to match the sole bottom.

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3.6.19 Upper finishing

3.6.19.1 Preparation. The boots shall be cleaned, removing accumulated dirt, wax, cement, and other foreign matter. All thread ends shall be trimmed.

NOTE: Upon occasion when some insignificant thread ends are noted during inspection (audit) they will not be cause for lot/shipment rejection. In-plant corrective action shall be made on current and future production.

3.6.19.2 Inserting molded cushion insole. An insole, as specified in 3.2.20, shall be inserted in each boot.

3.6.20 Final finish. The boots shall be repaired and filled. All raw edges, including vamp line, backstay and front edges shall be stained to match upper leather. An overcoat of a bright spray shall then be applied. Materials and methods shall be specified in 3.2.18 through 3.2.18.3.

3.6.21 Pairing and lacing. The boots shall be paired. A lace shall be put through the bottom eyelets of each boot, with the lace ends placed inside the boot.

3.6.22 Repairs of stitching

3.6.22.1 Upper fitting stitches. Repairs of types 301 and 401 stitches shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching one stitch in back of the end stitching.

b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item at fitting room release, shall be repaired by overstitching. The stitching shall start one stitch in back of the defective area and continue one stitch beyond the defective area onto the existing stitching. Repair stitching may not be done in a way that the waterproof bootie lining will be penetrated.

3.6.22.2 Inseaming and sole stitching. Repairs of inseaming and outsole stitches shall be in accordance with 3.6.22.1, a. and b., and in addition, thread cuts occurring as a result of trimming operations shall be repaired in the same manner as specified for thread breaks.

3.6.23 Heel assembly. In preparation for the heel to the outsole bond, the bevel of the heel shall be wire brushed to remove any dust and to ensure a good bond. The heel shall then be primed on both sides including the bevel with an adhesive specified in 3.2.17 and allowed ample time to dry. After drying the heel shall be cemented with an adhesive specified in 3.2.17 and allowed ample time to dry. The heel shall be spotted to the sole using a flash activator at

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125°F. The outsole/heel unit shall be placed in a sole laying machine with sufficient pressure to ensure a good bond.

3.7 Leakage. The finished boots shall be tested for leakage as specified in 4.4.4.

3.8 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects shall be removed from production and replaced with nondefective and properly matched components.

3.9 Workmanship. The end item shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests as specified herein). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and tested for the characteristics specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the characteristics listed in Table X conform to the specified requirements.

TABLE X. Component tests

Material	Characteristic	Paragraph	Test Method
Insoles	Material identification	3.2.2	-
Laminated cloth	Weight (oz./sq. yd.)	3.2.3.3	5041 2/
	Thickness (inches)	3.2.3.3	5030 2/
	Thermal shrinkage machine (%):		
	Sample mean	3.2.3.3	4.5.5
	Thermal shrinkage transverse (%):		
	Sample mean	3.2.3.3	4.5.5
	Moisture vapor transmission rate (g/m^2/24 hr.):		
	Procedure B	3.2.3.3	E 96 1/ 4/
	Procedure BW	3.2.3.3	E 96 1/ 5/

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TABLE X. Component tests (contd.)

Material	Characteristic	Paragraph	Test Method
	Water permeability:		
	Initial	no leakage	6/
	After aircraft fluid	no leakage	7/
	Colorfastness to:		
	Water	3.2.3.3.3	AATCC 107
	Perspiration	3.2.3.3.3	AATCC 15
	Crocking	3.2.3.3.3	AATCC 8
Laminated tape	Material identification	3.2.4.1	-
Film tape	Material identification	3.2.4.2	-
Collar foam	Material identification	3.2.5	-
Thermal insulation	Material identification	3.2.6	-
	Thickness	3.2.6	3/
Ribbing fabric	Material identification	3.2.7.1	-
	Shear strength	3.2.7.2	4.5.4
	Stitch strength	3.2 7.2	4.5.4
Box toe	Material identification	3.2.8	-
	Thickness	3.2.8	-
Shank fiberglass	Material identification	3.2.11	-
	Width	3.2.11	-
Staples	Material identification	3.2.12.1	-
Eyelets	Material identification	3.2.12.2	-
Speed lace loops	Material identification	3.2.12.3	-
	Finish	3.2.12.3	-
Bottom filler	Material identification	3.2.13.1	-
Outersole	Abrasive index	3.2.14.3	D 1630 1/
	Hardness	3.2.14.3	D 2240 1/

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TABLE X. Component tests (contd.)

Material	Characteristic	Paragraph	Test Method
	Volume swell	3.2.14.3	6001 and 6211 8/
	Split tear	3.2.14.3	FIA 324 9/
	Stitch tear	3.2.14.3	FIA 320 9/
	C tear	3.2.14.3	D 624 1/
	Ross flex	3.2.14.3	D 1052 1/
Adhesive	Material identification	3.2.17	-
	Bonding strength Midsole to outsole	3.2.14.4	D 816 (method B) (see 4.5.2)
Finishing	Compatibility	3.2.18	-
Insole, cushion	Material identification	3.2.20	-
	Thickness	3.2.20	-
	Density	3.2.20	D 3574 (Test A)1/
	Tensile strength	3.2.20	D 3574 (Test E)1/
	Compression set	3.2.20	D 3574 (Test D)1/
	Compression set after hydrolysis	3.2.20	D 3574 (Test D)1/
Bootie	Waterproofness	3.6.5	4.5.1

1/ Refers to ASTM Standards.

2/ Refers to FED-STD-191.

3/ Specimen size: 3.5 inch diameter, total load; 0.25 psi, loading time: 5 seconds.

4/ The back side of the laminate cloth shall face the water, the free stream air velocity shall be 550 ± 50 FPM as measured 2 inches above the fabric specimen. The air flow shall be measured at least 2 inches from any other surface. The test shall be run for 24 hours and weight measurement shall be taken only at the start and completion of the test. At the start of the 24 hour test period, the air gap between the water surface and the back of the specimen shall be 3/4 ± 1/16 inch. Five specimens shall be tested. The test chamber shall be 73.4^{o} ± $1^{o}F$ and the relative humidity shall be 50 ± 2 percent.

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5/ The back side of the laminate cloth shall face the water, the free stream air velocity shall be 550 ± 50 FPM as measured 2 inches below the fabric specimen. The air flow shall be measured at least 2 inches from any other surface. The test shall be run for 2 hours and weight measurements shall be taken only at the start and completion of the test. Five specimens shall be tested. The specimen shall be sealed in any manner which prevents wicking and/or leaking of water out of the cup. The test chamber shall be 73.5° ± 1°F and relative humidity shall be 50 ± 2 percent.

6/ The water permeability shall be measured as specified in Method 5516 of FED-STD-191, except that the back side of the laminated cloth shall contact the water. The hydrostatic head shall be 50 centimeters and shall be held for five minutes. Leakage is defined as the appearance of water any place within a 4-1/2 inch diameter test area. The test may be performed using any device which tests the same specimen area at the equivalent pressure. In cases of dispute, the apparatus described in method 5516 of FED-STD-191 shall be used.

7/ One specimen per sample unit shall be tested for leakage after exposure to aircraft fluids. The specimens shall not be less than 6 inches in diameter. Specimens shall be placed on a flat surface and 1 ML of JP-4 (jet fuel conforming to MIL-T-5624) spread over the middle of the back side, followed by 1 ML of deicing fluid conforming to MIL-A-8243. Place the horizontal specimens flat in an oven with air circulating over at 50°C for 30 minutes. Remove from oven and test immediately as specified in footnote 6/.

8/ Refers to FED-STD-601.

9/ Refers to Footwear Industries of American Standards.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 In-process examination. The defects found during in-process examination shall be classified in accordance with 4.4.2.1.1, 4.4.2.1.2 and 4.4.2.1.3. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major and 6.5 for total (major and minor combined) defects.

NOTE: Defects designated by an asterisk in tables XI through XIV shall be scored as major when seriously affecting serviceability and minor when affecting serviceability but not seriously.

4.4.2.1.1 Pre-fitting examination of cut parts. The cut parts shall be examined for the defects listed in table XI before fitting. The lot size shall be expressed in units of cut parts. The sample unit shall be one cut part. Defects shall be major when seriously

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affecting serviceability or appearance, and minor when affecting serviceability or appearance, but not seriously.

TABLE XI. Cut parts defects

Examine	Defect
Upper leather	Not full grain Leather deeply snuffed; i.e., fiber structure damaged Any grub or tick damage, scratches, brands, or boney leather Fat wrinkles or veins Stretchy vamp Leather embossed or printed
Gusset-tongue and collar leather	Flanky Loose flesh, boney, or boardy
Quality of leather	Thickness exceeding the maximum or less than the minimum specified Rough fiber on flesh side Off-stretch cut Slaughter cut

4.4.2.1.2 Examination of uppers after all fitting. The upper assemblies shall be examined for the defects listed in Table XII. The lot size shall be expressed in units of upper assemblies prepared for lasting. The sample unit shall be one completely fabricated upper assembly prepared for lasting.

TABLE XII. Uppers after all fitting defects

Examine	Defect	Classification Major	Minor
Construction and workmanship (general)	Any component missing or not specified type	101	
	Any component misplaced or not affixed as specified	*	
	Any cut, tear, hole, repair, or factory damage	*	
	Backseam not rubbed down		201

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TABLE XII. Uppers after all fitting defects (contd.)

Examine	Defect	Classification Major	Minor
Bootie lining	Torn, loose, or wrinkled		*
	Not properly mated with upper		*
Quality of leather	Thickness more than 1/2 ounce less than minimum specified	102	
	Thickness less than specified minimum but not extending 1/2 ounce less		202
	Thickness more than maximum specified		203
	Off-stretch cut		*
	Slaughter cut		*

4.4.2.1.3 In process examination of boot before bottom filling. The partially fabricated boots shall be examined for the defects listed in Table XIII. The lot size shall be expressed in units of partially fabricated boots. The sample unit shall be one partially constructed boot assembled to the point after lasting and attachment of shank, but before the application of the bottom filler.

TABLE XIII. Boot before bottom filling defects

Examine	Defect	Classification Major	Minor
Bottom of boot	Any component missing or not specified type	101	
	Shank not properly positioned or not fitting contour of boot bottom	102	
	Shank wrong size, malformed, or not securely attached	103	
	Upper damaged		*
	Poor heel seat, side, or toe lasting		*
	Inseam not properly trimmed		201
	Inseam stitches broken, two or more consecutive skipped inseam stitches, or inseam stitches cut or damaged during trimming operation		

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TABLE XIII. Boot before bottom filling defects (contd.)

Examine	Defect	Classification Major	Minor
	not repaired	104	
	not repaired as specified		202
	inseam stitches not in insole rib		203
	End of counter not caught in inseam stitching	*	
Upper part of boot	Upper not tightly pulled down to last	*	
	Quarters not laced as specified		204
	Upper in lasted area not sufficiently roughed or cleaned after roughing	105	
	Excessive roughing or scouring	*	

4.4.2.1.4 In process testing of finished boots for leakage. The fabricated boots shall be tested for leakage upon completion of inspection in the finishing department and prior to final testing. Fifteen percent of all items being manufactured shall be tested in accordance to 4.5.7. All defective boots shall be marked with proper identification and rejected.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in Table XIV. The lot size shall be expressed in units of boots. The sample unit shall be one completely fabricated boot and the selection shall be by pairs. The inspector shall check to see that insoles are inserted in the boots. For the pairing examination, and when determining possible differences in outersole thickness, the pair shall be examined together. Defects of pairing shall be classified as a single defect. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects. During the inspection for the defects listed within the "End item visual defect table", the departures from specification requirements listed in the table shall be observed and recorded. Presence of these departures shall not result in the rejection of the lot, but the government quality assurance personnel reserve the right to establish and impose lot rejection criteria when either the frequency or severity of these departures warrant. This additional rejection criteria may be invoked at any time during the life of the contract when the government judges this action to be in their best interests.

NOTE: Appearance defects shall be scored only when the condition is plainly visible at a distance of 3 feet or more and it appreciably affects item appearance.

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TABLE XIV. End item visual defects

Examine	Defect	Classification Major	Departure	
Pairing	Not properly mated; i.e., not right and left of same size	101		
	Variation in color, luster, or appearance		*	
	Variation of more than 1/4 inch in height of pair	102		
	Box toe crooked, long or short		*	
	Difference in outersole thickness between left and right boots:			
	- more than 3/32 inch	103		
	- more than 1/16 inch but not more than 3/32 inch			201
Cleanness	Any nonremovable spot, stain, or foreign matter affecting appearance			202
Color and finish	Color not within the specified range		*	
	Finish streaky, chipped, or flaky on upper			203
	Any raw edges not stained to match upper leather			204
Design, type	Not as specified	105		
Construction and workmanship (general)	Any cut, tear, hole, repair or factory damage		*	
	Any component or assembly omitted or misplaced, operation omitted, or not properly performed (unless otherwise classified herein)		*	
Seams and stitching (upper)	Open seam not repaired	106		
	Repair of open seam not as specified			205

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TABLE XIV. End item visual defects (contd.)

Examine	Defect	Classification Major	Classification Departure
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur. On multiple stitched seams, a seam is considered open when either one or both sides of the seam are open. When the above conditions occur on the inside of the boot it shall be scored as a * defect.		
	Tight tension resulting in puckering or cutting of the leather		*
	Loose tension resulting in a loosely secured seam	107	
	Wrong stitch type		*
	Stitching omitted where required		*
	Gage of stitching not as specified or irregular		206
	Thread ends not trimmed throughout the boot		*
	Needle holes or needle chew		
Vamp stitching at outersole junction	One or more rows of defective stitching not repaired	108	
	Repaired but first stitch more than 1/4 inch from outersole junction	109	
	Repaired with one stitch in lieu of two or more stitches		207
Outersole and heel	Thickness not as specified	110	
	Edge scouring is irregular, affecting appearance		208
	Blisters:		
	- 1/2 inch or more in longest dimension	111	
	- penetrating 1/3 or more the thickness of rubber	112	
	- more than three blisters, more than 1/8 inch but less than 1/2 inch in longest dimension	113	

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TABLE XIV. End item visual defects (contd.)

Examine	Defect	Classification Major	Departure
Outersole and heel (contd)	- one, two or three blisters, more than 1/8 inch but less than 1/2 inch in longest dimension		209
	Any other defect in tread or edge area	*	
	Incomplete or incorrect vulcanization of outersole or heel to uppers (i.e., rubber not securely attached)	*	
Eyelets	Number of eyelets not as specified but each row has the same number		210
	Not the same number of eyelets in each row	114	
	Eyelets not properly spaced within the row or misalignment between the rows to an extent interfering with proper lacing	*	
	Edge of any eyelet less than 1/16 inch from edge of quarter	115	
	Any eyelet not securely clinched		211
Speed laces	Number of speed lace loops not as specified but each row has the same number		212
	Not the same number of loops in each row	116	
	Loops not properly spaced within the row or misalignment between the rows to an extent interfering with proper lacing	*	
	Loops not perpendicular to edge of quarters or parallel to each other		213
	Any loop not securely riveted	117	
	Four splits in rivet clinch roll		214
	Five or more splits in rivet clinch roll	118	
	Distance from center of rivet to edge of quarter less than 7/32 or more than 9/32 inch	*	

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TABLE XIV. End item visual defects (contd.)

Examine	Defect	Classification Major	Departure
Marking and instruction tag	Missing, incomplete, incorrect, not applied in the specific manner, misplaced, illegible, or not specified size		215
Bootie lining	Torn, loose or wrinkled	*	
	Not properly mated with upper		216
Bar-code label/tag	Barcode omitted or not readable by scanner; human-readable interpretation (HRI) omitted or illegible, bar-coded typed not as specified; code density not as specified		
Cushion insole	Missing, wrong size, or wrong side up	119	

4.4.4 End item testing. The finished boot shall be tested for the characteristics listed in Table XV. All test results shall contain the individual values utilized in expressing the final results. For the leakage test, the sample size shall be 1% of the lot regardless of the lot size. The boot shall be unacceptable if two or more sample units fail the leakage test. As an alternate to the above sampling criteria, one boot may be tested from each days production (After Final Inspection) and if found acceptable, that days production can be considered acceptable. However, when a failure is noted; that days (packed) production shall be audited (sampled), as indicated in accordance with the above sampling procedures.

TABLE XV. End item tests

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as Pass or Fail	Results reported as Numerically to nearest
Leakage	3.8	4.5.3	1	X	--

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully shipped. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling
Content	Number per container is more or less than required

4.4.6. Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection

4.5.1 Bootie lining tests. The bootie will be tested for waterproofness before it is inserted into the boot using the test apparatus listed in 6.5.13. The test will be performed as follows:

a. Inflate the bootie with air to 1.0 psi gage static air pressure (measured inside the bootie) and clamp the top of the bootie to seal from air leakage.

b. Submerge inflated bottle in a water bath such that the toe of the bootie is at least 5 inches below the water's surface for 2 seconds. Remove from bath and wipe entire bootie surface with hand.

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c. Re-submerge inflated bootie to the same position for at least 10 seconds.

d. While submerged, the bootie shall be inspected for air leakage. The bootie fails if, at any single location on the exterior of the bootie, there is a discreet air bubble more than once every 7 seconds.

e. No more than two repairs shall be made on each bootie. All repairs shall be made with the seam sealing tape referenced in 3.2.4.1.

4.5.2 Bond strength test of outersole to midsole

4.5.2.1 Specimen. The specimen shall be a combined midsole-outsole unit which has aged at least 2 days. The outsole shall be separated from the midsole for a distance of approximately 2-1/2 inches from the toe end of the specimen.

4.5.2.2 Apparatus. A power-driven portable gage tong adhesion machine or an approved portable testing device of equal performance shall be used. The rate of travel of the power actuated grip shall be 2 inches per minute. The machine shall be operated without any device for maintaining maximum load indication.

4.5.2.3 Procedure. The separate toe ends of the specimen shall be clamped in the jaws of the machine, with the jaws 2 inches apart. The specimen shall extend outward at right angles to the direction of the application load. The machine shall be started and the outsole and midsole shall be pulled apart for a distance of 1 inch. At that instant, the load indicated on the machine shall be read and recorded. The load shall be divided by the width of the specimen at the corresponding line of separation.

4.5.3. Boot leakage test. Remove the ventilating or cushioned insert. Place the boot on a horizontal plane. Fill the boot with about 4 inches of water. Any boot displaying leakage after 2 hours shall be a test failure. The boots tested for leakage shall be thoroughly dried.

4.5.4 Testing of rib strength. The rib shall be tested for compliance with the requirements of 3.2.7.2. Method 2061 of FED-STD-311 shall be used for shear strength test, and Method 2171 of FED-STD-311 shall be used for stitch tear strength.

The sample unit shall be two ribbed insoles, one for each test, and the sample size shall be 5, regardless of the lot quantity. Requirements are applicable to the sample unit for individual unit requirements and lot average for average requirements. The lot shall be unacceptable if one or more sample units or the lot average of all units fail to meet the requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

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4.5.5 Thermal shrinkage resistance test. Three 15 inch by 15 inch specimens shall be cut from each sample unit. All specimens to be tested shall be conditioned in accordance with FED-STD-191 (Section 4, Atmospheric Conditions for Testing). Specimens shall be tested not more than 5 minutes after removal from conditioning. Specimen marking and measurements shall be in accordance with the procedure specified in AATCC Test Method 135. The forced circulating air oven shall achieve and maintain an air temperature of 181° +5, -0°F for a period of not less than 10 minutes. Oven recovery time after the door is closed shall not exceed 1 minute. The specimens shall be mounted on a suitable rack such that the specimens are suspended by two metal hooks near each top corner with 12 ± 5 inches separation between hooks. The entire specimen shall be exposed to the circulating air so that it is not less than 2 inches from any oven surface or other specimen, and so that the airflow is parallel to the plane of the material. The rack assembly shall be placed in the oven for 10 minutes. Specimen exposure time shall begin when the oven has recovered to an air temperature of 181° $+5^{\circ}$, -9°F. The specimens shall be conditioned prior to shrinkage measurement.

4.5.6 In process boot leakage test. Place a boot on a horizontal plane. Fill the boot with about 4 inches of water. Any boot displaying leakage after 1 hour shall be properly identified and rejected.

5. Packaging. Packaging shall be as specified by the procurement activity.

5.1 Label/tag. Each item shall be individually bar-coded with the type VIII, class 17 label/tag of DDD-L-20. This label/tag shall be located so that it is completely visible on the item when it is folded and/or packaged as specified and so that it causes no damage to the item.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The boots are intended for wear by military personnel of the Department of Defense.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Types, classes and sizes required (see 1.2.1).
c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article inspection is required, (see 3.1), the item will be tested and should be a first article sample. The contracting officer should include specific

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instructions in acquisition documents regarding arrangement for examinations, quantity, and testing and approval.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal or as directed by the contracting activity.

6.5 Source of supply

6.5.1. Leather

6.5.1.1 Style Thule J74000. A suggested source for the Style Thule J74000 leather specified in 3.2.1.1 is Volunteer Leather, Nashville, TN 37202.

6.5.1.2 Style Boulder HH 13339. A suggested source for the Boulder HH 13339 leather specified in 3.2.1.1 is S. B. Foot, Red Wing, MN 55066.

6.5.2 Insoles. A suggested source for the Texorist insole specified in 3.2.2 is USM Corporation, Texon Division, 1190 Huntington Road, P. O. Box 365, Russel, MA 01071.

6.5.3 Bootie lining material. A suggested source for the bootie lining material specified in 3.2.3.1 is W. L. Gore and Associates, 297 Blue Ball Road, Elkton, MD 21921.

6.5.4 Insulation.

6.5.4.1 Thinsulate. A suggested source for the Thinsulate B-200 and the Thinsulate B-400 is Thermal Insulation Products, 3M Center Building, 200-3E-04, St. Paul, MN 55144.

6.5.4.2 Thermolite. A suggested source for the Thermolite B-200 TFI and the Thermolite 400 TFI is Hobbs Bonded Fibers, 3010 LBJ Freeway, No 1207, Dallas TX. 75234.

6.5.5 Seam sealing tape

6.5.5.1 Seam sealing tape for bootie. A suggested source for tape as specified in 3.3.4.1 is W. L. Gore and Associates, 297 Blue Ball Road, Elkton, MD 21921.

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6.5.5.2 Seam sealing tape for heel and counter. A suggested source for the tape specified in 3.2.4.2 is Bemis Associates, Inc., One Bemis Way, Shirley, MA 01464-2527.

6.5.5.3 Seam sealing tape machine. A suggested source for the machine to apply the seam sealing tape as specified in 3.2.4.1 and 3.2.4.2 is W. L. Gore and Associates, 297 Blue Ball Road, Elkton, MD 21921.

6.5.6 Box toes. A suggested source for the box toes specified in 3.2.8 is Brockton Sole and Plastic, 53 Spark Street, P. O. Box 2128, Brockton, MA 02403.

6.5.7 Fiberglass shank. A suggested source for the fiberglass shank specified in 3.2.11 is American Shoe Machinery Co., (designated "TRU-FIT"), 199 Newbury Street, P. O. Box 390, Danvers, MA 01923.

6.5.8 Shankboard. A suggested source for the Bonfoam 300HD (see 3.2.11.2) is Georgia Fibers, Inc., Buena Vista, VA 24416.

6.5.9 Outersole. A suggested source for the outersole specified in 3.2.14.2 is Quabaug Corp., (designated "VIBRAM"), North Brookfield, MA 01535 or equal.

6.5.10 Molded cushion insoles. Suggested sources for the cushion insoles specified in 3.2.20 are Gemini Laminating Corporation (Product No. 11995-BC), 166 Forbes Road, Braintree, MA 02184. Frelonic (Product No. FT-9503), 63 Grove Street, P. O. Box 169, Salem, MA 01970 or Rogers Foam, 145 Central Street, Somerville, MA 02145.

6.5.10.1 Cushion midsole crepe. A suggested source for the cushion midsole crepe specified in 3.2.14.1.1 is Biltrite, #2 University Office Park, Waltham, MA 02254.

6.5.11 Microcellular foam. A suggested source for the microcellular foam meeting the requirements of 3.2.20 is Rogers Foam, Poron Division, East Woodstock, CT 06244.

6.5.11.1 Alternate cushion insole. Source for the alternate cushion insole meeting the requirements of 3.3.20.1 is Atlantic Thermoplastics, P.O. Box 457, Blackstone MA 01504.

6.5.12 Lasts

6.5.12.1 M2673B-1 lasts. A suggested source for the M2673B-1 lasts specified in 3.5 is Vulcan Corp.

6.5.12.2 701 lasts. A suggested source for the 701 lasts specified in 3.5 is Jones and Vining Corp.

6.5.13 Bootie lining testing apparatus. A suggested source for the testing apparatus needed for testing in 4.5.1 is W. L. Gore and Associates, 297 Blue Ball Road, Elkton, MD 21921.

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6.6 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO STANAG 2333. When amendment, revision, or cancellation of this document is proposed that will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization office, if required.

6.7 Equal item. Prior to use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.8 Subject term (key word) listing.

Footwear
Insulated
Leather, black
Liner
Waterproof

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU

Review activities:

Army - MD
Navy - MC

Preparing activity:
DLA-CT

Project No. 8430-0422

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INSTRUCTION TAG

BOOT, INTERMEDIATE, COLD/WET

FACTS

These Intermediate Cold/Wet Boots (ICWB) have been designed to protect your feet from cold injury and frostbite in areas where moisture and cold are critical factors, and where the mean monthly temperature ranges between 14°F and 68°F. The boots should not be worn where temperatures fall below +10°F. (For temperature below +10°F, the Cold Weather, Insulated, Rubber, Black or White, Vapor Barrier Boots should be worn.)

If your feet begin to get cold, button up all your garments and exercise the entire body by swinging the arms, stamping the feet, and rapid movement of the toes.

TRENCH FOOT AND FROSTBITE are serious cold injuries which can cause painful and permanent disability. These injuries can be prevented by proper care of your feet. No boot will keep your feet from freezing in cold/wet conditions if you remain motionless for several hours. Wear the ICWB properly and DO NOT BECOME A COLD WEATHER CASUALTY.

INSTRUCTIONS FOR USE

1. Wear with: 1 pair of socks, wool, cushion sole.

2. Sizes. The boots are supplied in whole and half sizes 3 through 14, and in widths, N, R, W, and XW. In some instances it may be necessary to select a slightly larger size than normally worn in order to allow for normal swelling of the feet.

3. Pull on the boots, seating heel firmly in place, then lace. Boots should fit snugly but not tightly. There should be approximately 1/2 inch additional length at toe.

4. Carry extra socks. Change socks often, at least once daily. Remove boots and dry feet as often as possible. Neglect of feet can make you a casualty.

5. When your boots become soiled or dirty, simply brush or wipe them clean with a damp cloth. Let them dry naturally at room temperature (do not dry next to radiator). Waterproofing is not required, however, use of conditioners such as mink oil will help to maintain the natural condition of the leather and extend the service life of the boots.

FIGURE 1. Instruction tag

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
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3. DOCUMENT TITLE
BOOTS, INTERMEDIATE COLD/WET

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY	
a. NAME Defense Personnel Support Center ATTN: DPSC-FQSX (12-3-C)	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)* 2800 South 20th Street Philadelphia, PA 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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